

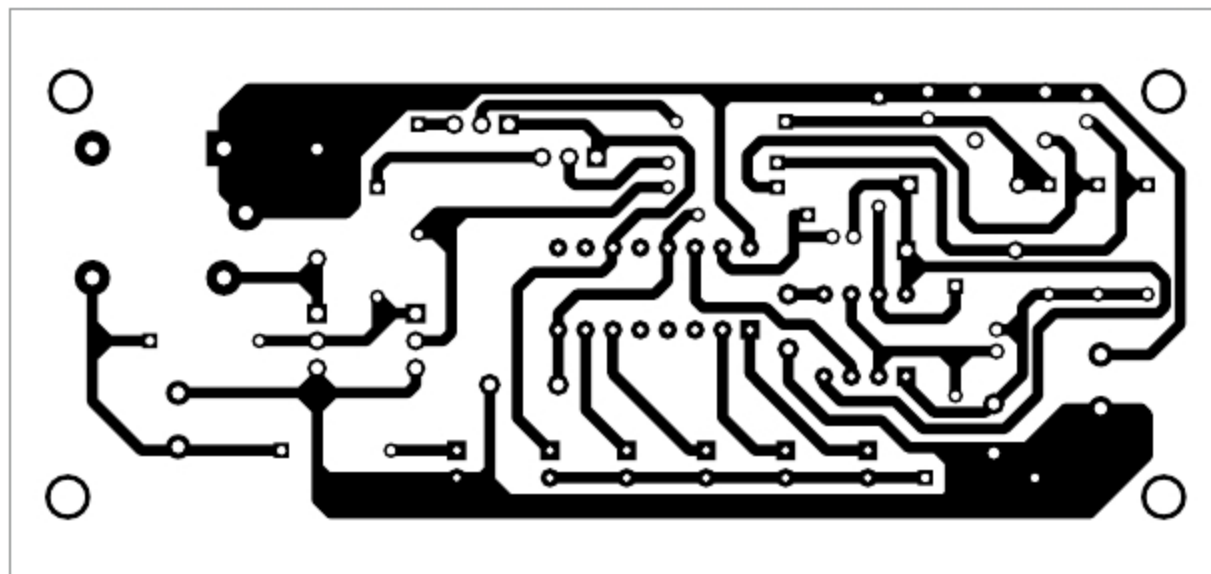


Fig. 2: Actual-size PCB layout of the protection circuit

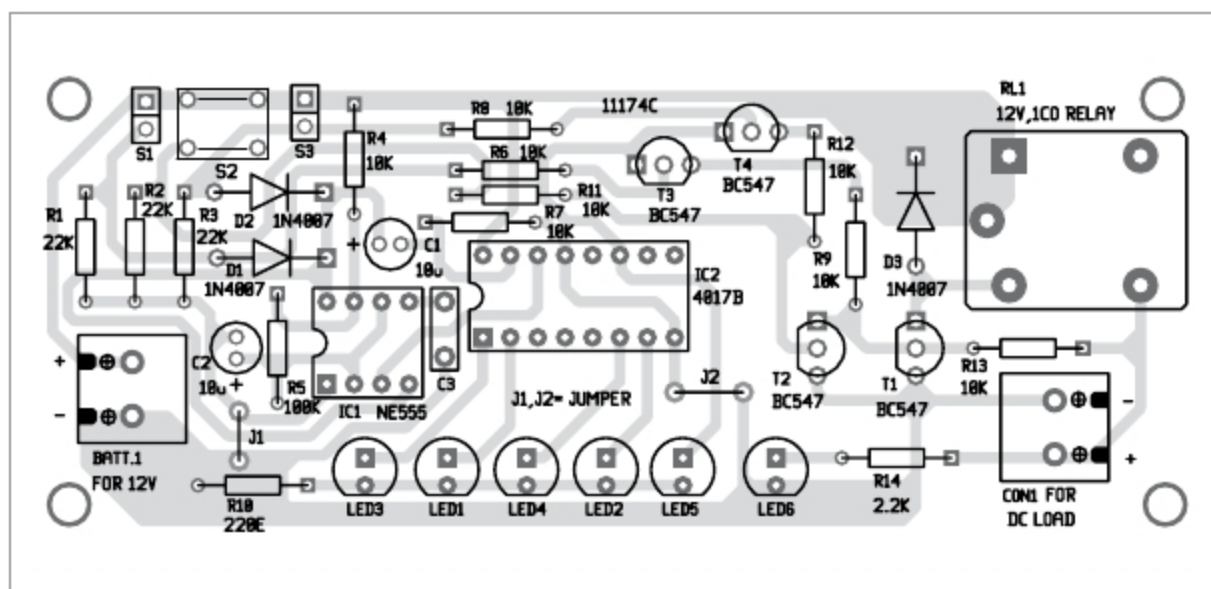


Fig. 3: Components layout for the PCB

PARTS LIST

Semiconductors:

- IC1 - NE555 timer
- IC2 - 4017B decade counter
- T1-T4 - BC547 npn transistor
- D1-D3 - 1N4007 rectifier diode
- LED1-LED6 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R3 - 22-kilo-ohm
- R4, R6-R9, R11-R13 - 10-kilo-ohm
- R5 - 100-kilo-ohm
- R10 - 220-ohm
- R14 - 2.2-kilo-ohm

Capacitors:

- C1, C2 - 10 μ F, 25V electrolytic
- C3 - 10nF ceramic disk

Miscellaneous:

- CON1 - 2-pin terminal connector
- S1, S3 - On/off switch
- S2 - Tactile switch
- BATT.1 - 12V battery
- RL1 - 12V, single-changeover relay
- 12V DC load

SWITCHES AND THEIR FUNCTIONS

Switch	Relay RL1	DC load
Close S1 for delayed on	Energised and latched	Delayed on and LED6 on
Press S2 momentarily for quick off	De-energised quickly	Quick off and LED6 off
Close S3 for delayed off	De-energised	Delayed off and LED6 off

layout in Fig. 3. After assembling the circuit on a PCB, connect 12V battery across BATT.1 and 12V DC load across CON1.

EFY note. To replace DC with AC load, you need another SPDT relay with its coil connected to existing relay output, or a single DPST/DPDT switch for replacing the existing relay (not shown in circuit diagram). **EFY**

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Construction and testing

An actual-size PCB layout for the accidental switching protection circuit is shown in Fig. 2 and its components